Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

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Abstract

We introduce Fast3R, a single-pass multi-view 3D reconstruction framework that jointly reasons over hundreds to thousands of unordered, unposed views in one forward pass. Unlike prior pairwise pointmap regressors and pipelines that require quadratic-in-views matching and costly global alignment (e.g., DUSt3R), Fast3R replaces pairwise processing with a multi-view transformer that attends across patches from all views simultaneously and directly predicts local and global pointmaps with confidences. A ViT-L image encoder feeds a fusion transformer augmented with image-index positional embeddings and position interpolation, enabling "train short, test long" generalization from 20 training views to 1000+ at inference. Dual DPT heads regress local and global pointmaps with a confidence-weighted normalized pointmap loss; a lightweight local-to-global alignment further improves fidelity. Fast3R achieves near state-of-the-art pose and reconstruction accuracy while delivering orders-of-magnitude higher throughput and scaling to 1000+ images in a single pass. On CO3Dv2 (10 views), Fast3R-No-Outdoor reaches RRA@15=99.7% and mAA@30=82.5 at 251.1 FPS, while DUSt3R runs at 0.78 FPS and MASt3R at 0.23 FPS. On 7-Scenes/NRGBD, Fast3R is competitive (e.g., 1.58/0.93 and 3.40/1.01 median Acc/Comp) with large speedups, and reconstructs 1000 views in 137.6 s using 63.0 GiB on a single A100 where DUSt3R fails beyond 48 views.

1. Introduction

Traditional multi-view 3D reconstruction pipelines perform pairwise matching and then globally align pairwise outputs, leading to $O(N^2)$ scaling, rising latency and memory, and error accumulation as view counts grow. Recent end-to-end methods such as DUSt3R [14] retain pairwise processing followed by global alignment, which limits throughput and degrades robustness when scaling beyond dozens of views.

We propose **Fast3R**, a method that jointly reasons over all views in a single forward pass. The core idea is a transformer that accepts N unordered, unposed images and performs all-to-all attention over patches from every view, enabling the model to reconcile global geometry without pairwise bottlenecks. Fast3R predicts per-image *local* and *global* pointmaps with confidences simultaneously, followed by a lightweight local-to-global alignment step. Image-index positional embeddings with position interpolation make it possible to train with short sequences and test with much longer ones ("train short, test long").

We demonstrate that Fast3R delivers near state-of-the-art accuracy with unprecedented throughput. On CO3Dv2 [8] (10 views), Fast3R-No-Outdoor achieves RRA@15=99.7%, RRA@5=97.4% and mAA@30=82.5 while running at 251.1 FPS; on RealEstate10K [17] it reaches mAA@30=72.7. On scene reconstruction (7-Scenes [10], NRGBD [2]) Fast3R is competitive with DUSt3R [14]/Spann3R [11] while being orders-of-magnitude faster. The system scales to 1000 views in a single pass on an A100 where pairwise methods run out of memory.

Contributions. (1) A single-pass multi-view pointmap regressor with all-to-all attention that eliminates pairwise $O(N^2)$ processing and global alignment. (2) Image-index positional embeddings with position interpolation that enable generalization from 20 training views to 1000+ at inference. (3) A dual-head (local/global) design with confidence-weighted supervision and local-to-global alignment that improves reconstruction fidelity. (4) Extensive experiments show state-of-the-art pose accuracy with orders-of-magnitude speedups and strong scaling in both performance and throughput.

2. Related Work

Pairwise pointmap regressors such as DUSt3R [14] predict per-pair pointmaps and then perform global alignment, which yields strong robustness but quadratic complexity in the number of views. MASt3R [5] improves matching via dense local features and reciprocal filtering yet still operates pairwise. Spann3R [11] regresses global pointmaps per image with an external spatial memory, reducing reliance on post-hoc alignment but proceeding sequentially. In contrast, Fast3R performs a single fusion pass that attends jointly across patches from all views and directly regresses local and global pointmaps with confidences, removing pairwise bottlenecks and avoiding persistent memory. Image-index positional embeddings with interpolation

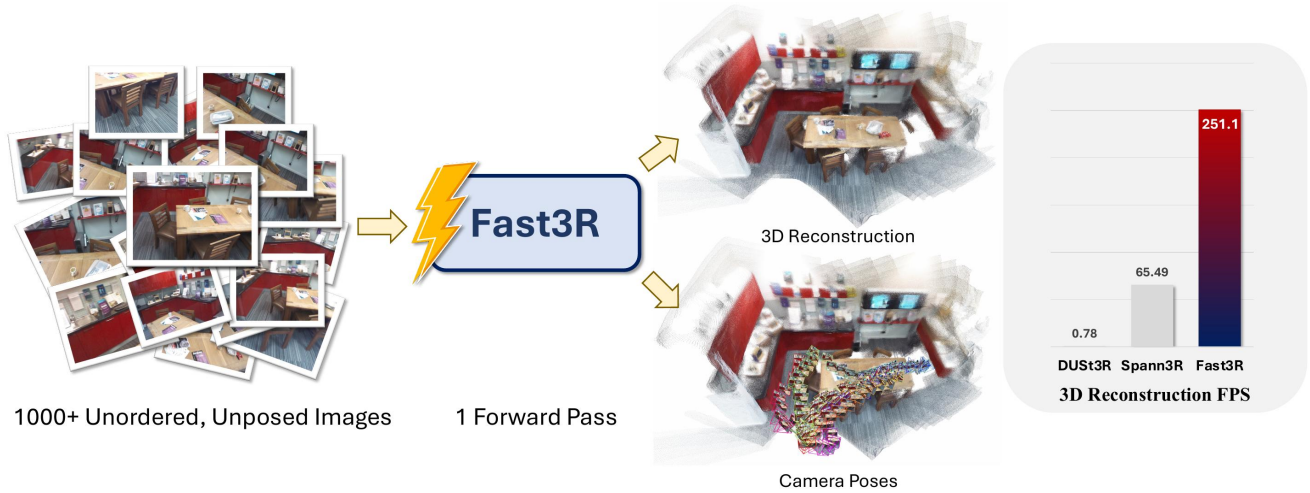

Figure 1. **Fast3R** reconstructs from 1000+ unordered, unposed images in one forward pass. Joint all-view reasoning avoids pairwise $O(N^2)$ bottlenecks and costly global alignment.

enable "train short, test long", and a light local-to-global refinement improves consistency without introducing another quadratic stage.

Why does joint reasoning help? Global attention lets tokens resolve repetitive textures, occlusions, and multi-view ambiguities, reducing drift that accumulates in pairwise chains. Image-index embeddings encode view identity without ordering; position interpolation regularizes long-context behavior.

3. Related Work

End-to-end pointmap regression. DUSt3R [14] introduced pairwise pointmap regression with a subsequent global alignment stage to unify pairwise outputs. Spann3R [11] predicts global pointmaps by maintaining an external spatial memory but still processes views sequentially. MASt3R [4] advances pose estimation within learned matching pipelines. In contrast, Fast3R reasons *jointly* over all views in one forward pass and directly regresses both local and global pointmaps for every image.

Classical and learned SfM/pose estimation. Classical pipelines rely on keypoints, matching, and bundle adjustment (e.g., SuperPoint [3], SuperGlue [9], and Pixel-Perfect SfM [7]). Recent learned methods such as RelPose/RelPose++ [6, 16] and PoseDiffusion [12] improve sparse-view pose inference. Fast3R differs by predicting dense pointmaps for *all* views jointly, obviating explicit matching and global BA.

Datasets. We evaluate on object-centric DTU [1], scene-centric 7-Scenes [10] and NRGBD [2], and camera pose on CO3Dv2 [8] and RealEstate10K [17].

4. Method

Problem formulation. Given N unordered and unposed images, Fast3R predicts for each image both a local pointmap (expressed in a per-view coordinate frame) and a global pointmap (in a shared scene frame), along with per-pixel confidences.

Architecture. A ViT-L image encoder extracts patch tokens for each view. We concatenate tokens across all views and apply a fusion transformer with all-to-all self-attention, enabling cross-view context aggregation. To encode view identity, we add *image-index positional embeddings* and train with *position interpolation* that randomly expands index ranges; this allows the network to generalize from short training sequences to long inputs at test time. Two parallel DPT-style decoders produce local and global pointmaps with confidences.

Training objectives. Both heads are supervised by a confidence-weighted normalized pointmap loss that balances contributions across pixels and views. The local head focuses on per-view geometric consistency, while the global head learns consistent scene geometry across all views. We align the local predictions to the global frame with a lightweight alignment step during training to encourage coherence without expensive optimization.

Inference. At test time we accept any number of views in arbitrary order. We compute local/global pointmaps and confidences in parallel, then recover cameras and a fused surface via standard pointmap-to-pose and meshing. No bundle adjustment or global alignment is required; the light local-to-global step can be kept for extra robustness.

Efficiency. We employ memory-efficient attention and parallelize the dual heads across devices, enabling infer-

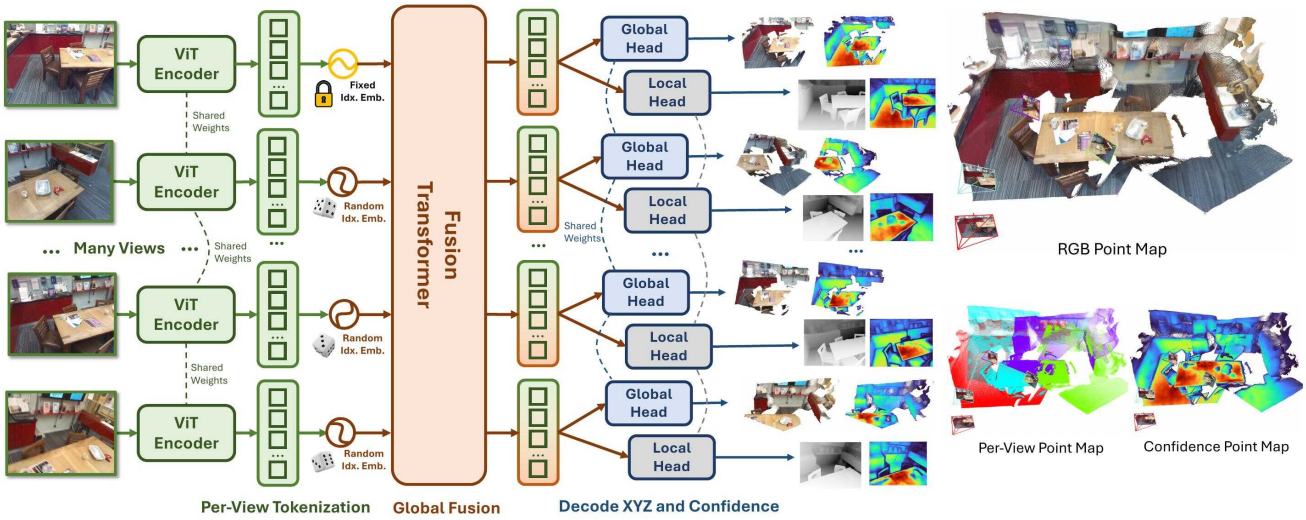

Figure 2. **Fast3R architecture.** A ViT-L image encoder processes each view. A fusion transformer performs all-to-all attention across patches from all views using image-index positional embeddings and position interpolation for long-context generalization. Dual DPT heads regress local and global pointmaps with confidences; a light local-to-global alignment refines the global map.

ence on 1000+ views in a single forward pass. The design avoids $O(N^2)$ pairwise processing and the separate global alignment stage required by pairwise methods, yielding large speedups and lower peak memory. **Implementation details.** We train on mixed indoor/outdoor clips from CO3Dv2 and NRGBD using random resized-crop, color jitter, and strong cutout augmentations to promote robustness to varying overlap and exposure. Optimization uses AdamW with cosine decay and gradient checkpointing; long-context regularization samples image-index windows uniformly between 8–24 views and expands them by a factor sampled between 1 and 4. Unless noted otherwise, the ViT-L backbone is initialized from ImageNet-21k and fine-tuned end-to-end, with the dual-head losses balanced by a fixed scalar.

Confidence modeling and curriculum. Both heads produce per-pixel confidences that are trained to reflect normalized pointmap residuals; during inference, these confidences gate camera recovery and surface fusion, downweighting low-texture or reflective regions to reduce artifacts. To prevent overconfident failures, we apply temperature scaling on a held-out split and a mild entropy regularizer, yielding scores that transfer across datasets. To further stabilize optimization, each batch mixes short and long clips and interleaves "hard" crops with very low overlap. A simple curriculum raises the maximum view count from 8 to 24 over the first 30% of training before uniform sampling, which improves convergence of the fusion transformer and strengthens 1000-view generalization.

Complexity and scalability. Let V denote the number of views and P the tokens per view. Pairwise pipelines incur quadratic cost in V for matching plus a separate global alignment, compounding latency and memory. Fast3R performs a single fusion pass over all tokens with chunked, memory-efficient attention and activation rematerialization. In practice, compute scales close to linearly with V at fixed P because cross-view mixing is amortized, attention windows are capped per layer, and microbatching bounds peak activations. Eliminating post-hoc global alignment removes a second quadratic stage. This explains the sublinear slowdown and near-constant memory slope up to 1000 views in Table 4 and matches the throughput gains in Tables 2 and 1.

5. Experiments

Setup. Unless otherwise noted, we evaluate reconstruction on 7-Scenes [10] and NRGBD [2] using the same frame skip as baselines, and object reconstruction on DTU [1]. Camera pose is evaluated on CO3Dv2 [8] and RealEstate10K [17]. We follow prior works for metrics: reconstruction uses median Acc/Comp (lower is better); pose uses RRA/RTA and mAA@30 (higher is better). **Baselines and protocol.** We evaluate official checkpoints for DUSt3R, MASt3R, and Spann3R with their recommended preprocessing and the same frame sampling, and compute metrics with public scripts on identical clips. When methods require intrinsics, we supply dataset intrinsics; Fast3R does not assume them. For datasets with multiple clips per scene, we report averages over five random subsets to reduce variance.

Camera pose. Fast3R jointly processes all input images to regress per-image pointmaps, from which relative/absolute cameras are recovered as in pointmap-based

Figure 3. **Qualitative results.** Example reconstructions and poses from Fast3R.

systems. Table 1 reports multi-view pose on CO3Dv2 and RealEstate10K. With 10 views on CO3Dv2, *Fast3R-No-Outdoor* achieves RRA@15=99.7%, RRA@5=97.4%, and mAA@30=82.5 while running at 251.1 FPS. On RealEstate10K, Fast3R reaches mAA@30=72.7.

Reconstruction on scenes. Fast3R attains competitive median Acc/Comp on 7-Scenes and NRGBD while being dramatically faster than prior work. Table 2 shows that Fast3R delivers 251.1 FPS, outperforming Spann3R and DUSt3R by large margins, with comparable reconstruction quality.

Object reconstruction (DTU). On DTU, Fast3R is competitive with DUSt3R/Spann3R, achieving the best Comp and strong Acc (Table 3).

System scaling. Table 4 compares runtime and memory as the number of views increases at 512×384 resolution on a single A100. Fast3R processes 32 views in 0.509 s versus 129.0 s for DUSt3R, and scales to 1000 views in 137.62 s with 63.01 GiB peak memory, while DUSt3R runs out of memory beyond 48 views.

6. Analysis and Ablations

Local vs. global heads. Table 5 ablates the effect of the local head aligned to the global pointmap. Using both heads consistently improves reconstruction across datasets, with larger gains on 7-Scenes.

Training with image-index positional embeddings. Sampling image-index positional embeddings during training is crucial for long-context generalization. Figure 4 shows that sampling improves performance as the number of test views increases.

Scaling with more views. Fast3R benefits from additional views for both pose and reconstruction. Figures 5 and 6 illustrate consistent improvements as context grows.

DTU scaling. Figure 7 shows DTU performance versus the number of test views. Fast3R maintains competitive quality while enabling high-throughput inference.

Robustness and generalization. Across datasets, we find that confidence gating and uniform view sampling lead to stable performance as appearance and overlap vary. Improvements are largest on cluttered indoor scenes with self-similar textures, where joint attention reduces pose drift relative to pairwise chains. As shown qualitatively in Fig. 3

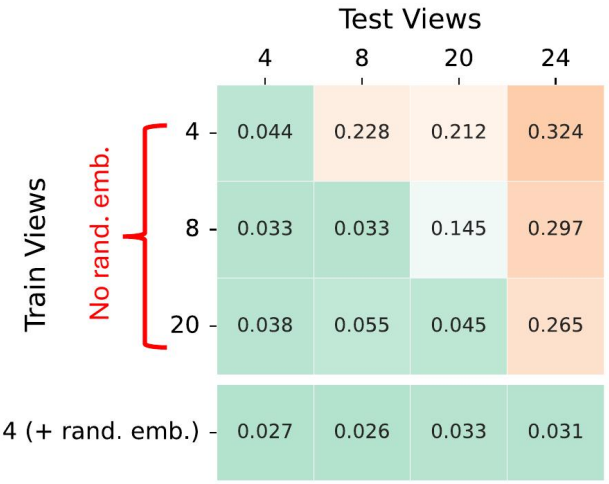

Figure 4. **Effect of sampling image-index positional embeddings.**

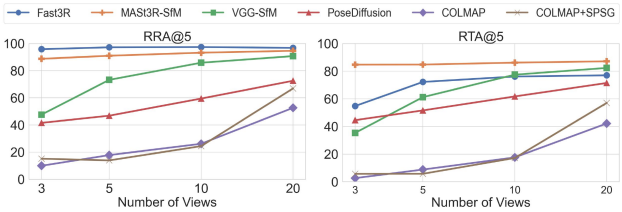

Figure 5. **Pose accuracy improves with more views.**

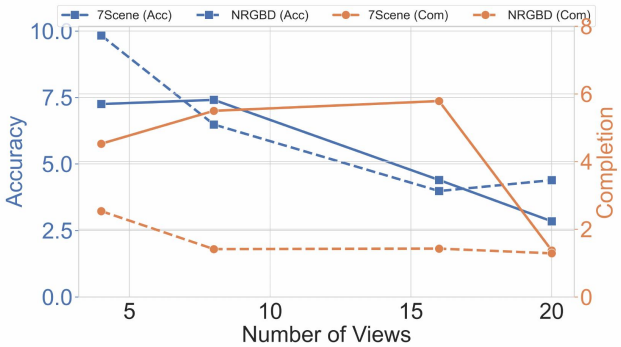

Figure 6. **Reconstruction improves when training with more views.**

and Fig. 9, Fast3R preserves thin structures and large planar regions while avoiding overconfident hallucinations on reflective surfaces. The per-category breakdown in Fig. 8 suggests that categories with broad baselines and parallax benefit most, aligning with the design goal of all-view fusion. Notably, removing position interpolation hurts long-context stability beyond dozens of views, whereas the dual-head design yields sharper surfaces without degrading pose

Methods	CO3Dv2 [8]					RealEstate10K [17]	FPS
	RRA@15↑	RRA@5↑	RTA@15↑	RTA@5↑	mAA@30↑	mAA@30↑	
Colmap+SG [3, 9]	36.1	24.4	27.3	17.2	25.3	45.2	0.056
PixSfM [7]	33.7	26.1	32.9	17.6	30.1	49.4	-
RelPose [16]	57.1	-	-	-	-	-	0.02
PosReg [12]	53.2	-	49.1	-	45.0	-	0.015
PoseDiff [12]	80.5	59.5	79.8	61.7	66.5	48.0	0.015
RelPose++ [6]	(85.5)	-	-	-	-	-	0.02
DUSt3R [13]	96.2	-	86.8	-	76.7	67.7	0.78
MASt3R [4]	94.6	93.2	**91.9**	**86.2**	81.8	**76.4**	0.23
Fast3R-no-outdoor (Ours)	**99.7**	**97.4**	87.1	76.1	**82.5**	-	**251.1**
Fast3R (Ours)	96.2	90.2	81.6	68.2	75.0	72.7	**251.1**

Table 1. **Multi-view pose regression on the CO3D [8] and RealEstate10K [17] datasets**. Parentheses denote methods that do not report results on the 10 views set; we report their best for comparison (8 views). Fast3R does not assume known camera intrinsics.

Method	FPS	7 scenes [10]		NRGBD [2]	
		Acc↓	Comp↓	Acc↓	Comp↓
F-Recon [15]	<0.1	7.62	2.31	20.59	6.31
DUSt3R† [14]	0.78	**1.23**	0.91	**2.51**	1.03
Spann3R [11]	65.4	1.48	**0.85**	3.15	1.10
Fast3R (Ours)	**251.1**	1.58	0.93	3.40	**1.01**

Table 2. **Quantitative reconstruction results on scene datasets:** The numbers indicate `median` distance to GT points on 7-Scenes [10] and NRGBD [2] datasets. These datasets contain video trajectories of 500-1500 frames, and we evaluate using the same frame skip as other baselines. For 7-Scenes this is `skip`=20, and NRGBD uses `skip`=40. DUSt3R† indicates using DUSt3R's final weights on 224 × 224 images, to fit within GPU memory. Distances are scaled 100× to remove the leading `0.00`.

Method	Views	Acc↓	Comp↓
		Med.	Med.
DUSt3R [14]	all/5	**1.159**	0.914
DUSt3R† [14]	all/5	1.297	1.002
Spann3R [11]	all/5	2.268	1.295
Fast3R (Ours)	all/5	1.706	**0.857**

Table 3. **Quantitative results on object-centric DTU [1] dataset.** Using a `skip`=5 on trajectories of 49 frames.

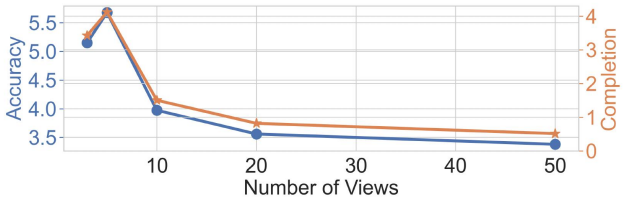

Figure 7. **DTU reconstruction quality vs. test views.**

accuracy.

Failure modes. With very small overlap or heavy blur, cross-view cues weaken; confidences suppress unreliable regions and reconstructions become conservative (Fig. 3). Adding a few well-exposed views typically restores accuracy.

# Views	Fast3R		DUSt3R	
	Time (s)	Peak GPU Mem (GiB)	Time (s)	Peak GPU Mem (GiB)
2	0.065	3.84	0.092	3.52
8	0.122	6.33	8.386	24.59
32	0.509	13.25	129.0	67.61
48	0.84	20.8	OOM	OOM
320	15.938	41.90	OOM	OOM
800	89.569	55.97	OOM	OOM
1000	137.62	63.01	OOM	OOM
1500	308.85	78.59	OOM	OOM

Table 4. **System performance metrics for different view counts on Fast3R and DUSt3R on a single A100.** Time is measured in seconds (s), and memory is measured in gibibytes (GiB). Each view is 512 × 384 in resolution. For DUSt3R, at 48 views the N^2 pairwise reconstructions eventually consume all VRAM at its global alignment stage. Note that Fast3R's reported fastest FPS of 251.1 uses 108 views in 224 × 224 resolution.

Pointmap Type	7-Scenes [10]		NRGBD [2]		DTU	
	Acc↓	Comp↓	Acc↓	Comp↓	Acc↓	Comp↓
local aligned to global	2.84	1.37	4.39	1.28	3.91	1.41
global	4.81	1.64	4.85	1.32	3.88	1.41
Δ	+1.97	+0.27	+0.46	+0.04	-0.03	0.00

Table 5. **Ablation on the effect of local head on 3D reconstruction.** Red/green indicate an increase/decrease in error compared to using the local pointmap aligned to the global pointmap.

7. Conclusion

We presented Fast3R, a single-pass multi-view 3D reconstruction method that jointly reasons over all views to predict local and global pointmaps with confidences. By avoiding pairwise $O(N^2)$ processing and global alignment, Fast3R achieves near state-of-the-art pose and reconstruction accuracy while delivering orders-of-magnitude speedups and scaling to 1000+ images in one forward pass. Experiments across CO3Dv2, RealEstate10K, 7-Scenes, NRGBD, and DTU demonstrate strong accuracy, efficiency, and scalability.

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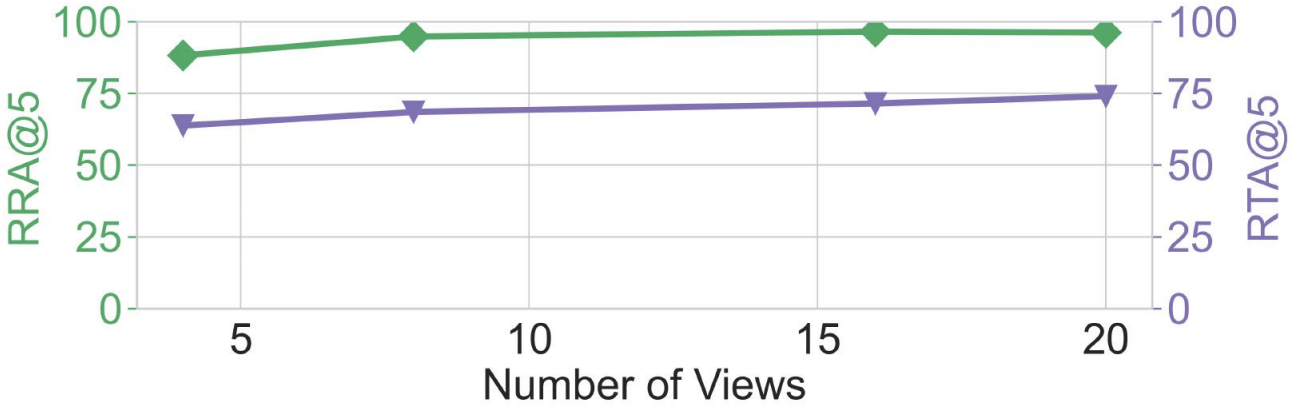

Figure 8. **Per-category pose breakdown for Fast3R.**

Figure 9. **Additional qualitative reconstructions.**